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ABSTRACT

In this study, the coupled dynamics of droplet impact on flexible superhydrophobic substrates are experimentally investigated using high-speed imaging and position sensing techniques across a range of Weber numbers ($We = 7.06\text{--}70.57$) and substrate flexibilities ($E = 0.45\text{--}1.40$ MPa). Regarding the droplet impact dynamics, the spreading phase is not significantly influenced by the Weber number or substrate flexibility, as inertial forces dominate during this stage, while the influence of substrate deformation remains relatively minor. In contrast, the receding phase is predominantly governed by the Weber number. With respect to the oscillation dynamics of the flexible substrate, the vibrational deflection increases with higher Weber numbers and lower substrate flexibility, whereas the vibration period is solely determined by the substrate's mechanical properties. The contact time observed on flexible substrates is longer than that on rigid substrates, which can be attributed to the scaling relationship between the oscillation periods of the droplet and the substrate. Based on these observations, a predictive correlation for the contact time of a droplet impacting flexible superhydrophobic substrates has been proposed. These findings provide valuable insights for the design of droplet-flexible substrate systems aimed at controlling contact time, offering a theoretical foundation for predicting system performance through the use of flexible materials.

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I. INTRODUCTION

Droplet impact on surfaces is a phenomenon commonly observed in practical applications, such as pesticide spraying,¹ inkjet printing,^{2,3} and anti-icing.^{4–6} During the impact phase, the droplet could undergo a spreading and receding phase, which may produce rebound, adhesion, splashing, and other phenomena. Most of the existing studies focus on the rebound phenomenon, which is more sensitive to interfacial physical parameters, i.e., surface wettability,⁷ micro- and nano-structural roughness,^{8,9} and surface chemistry¹⁰ that can all determine the critical condition for rebound. In industrial applications, the design of anti-icing coatings requires the suppression of harmful rebound, and the precise positioning of inkjet prints involves the control of rebound to avoid the generation of satellite drops.

The rebound phenomenon is marked by the complete detachment of the droplet from the surface, and the contact time (which is defined as the time between the droplet's first contact with the surface and its complete detachment) is a direct reflection of the system's dynamic equilibrium in terms of energy dissipation and the interactions between the droplet and the surface.¹¹ The contact time connects the spreading phase (conversion of kinetic energy into surface energy)

and the receding phase (conversion of surface energy into kinetic energy).¹² If the surface tension dominated restoring force during the receding phase is sufficient to overcome the viscous drag and adhesion energy, the droplet can be disengaged from the surface (the contact time is finite);^{13,14} if the energy dissipation is too strong (e.g., highly viscous liquids or strongly adhering surfaces), the droplet may end up adhering (the contact time tends to infinity).¹⁵ Therefore, the contact time visualizes the critical conditions for rebound, and its variation pattern is highly coupled with the rebound dynamics.

Currently, the contact time has been extensively investigated. Richard *et al.*¹⁶ first measured the contact time of a droplet on a superhydrophobic surface and found that the contact time can be expressed as $t_c \sim (\rho R_0^3 / \sigma)^{1/2}$, where ρ is the density, σ is the surface tension, and R_0 is the initial radius of the droplet. They also pointed out that the contact time is mainly determined by the balance of inertial force and surface tension of the droplet, which is independent of the impact velocity on a rigid superhydrophobic surface. Liu *et al.*¹⁷ designed a superhydrophobic surface with specific microstructures (e.g., arrays of conical tables) that enable “pie rebound” upon droplet impact, reducing the contact time by about 80%. The core mechanism is that

microstructure-induced capillary forces drive asymmetric contraction of the droplets, thereby accelerating disengagement. Lathia *et al.*¹⁸ identified two mechanisms for reducing contact time when a droplet with surface-coated hydrophobic particles impacted a superhydrophobic surface: reduced adhesion at low Weber numbers (21% reduction in contact time) and particle-induced fragmentation at high Weber numbers (65% reduction in contact time). A quantitative relationship between particle size, coverage, and contact time is revealed. The previous studies have revealed the mechanisms regulating the contact time of droplet impacts on rigid superhydrophobic surfaces from theoretical foundations, surface design (micro- and nanostructures, macroscopic ridges) to multi-physical field coupling (particle coatings). However, these studies are primarily confined to rigid surfaces and seldom address the influence of surface flexibility. Flexible surfaces possess broader practical applications, including prolonging the retention time of pesticide droplets during spraying processes, enhancing the anti-icing and anti-fogging performance of flexible sensors, and improving the operational stability of unmanned aerial vehicles under rainy conditions.^{19–22} Further investigation into the interaction between liquid droplets and flexible substrates is anticipated to yield significant advancements in both efficiency and functionality. The primary distinction between flexible and rigid surfaces lies in the pronounced deformation exhibited upon droplet impact. This deformation induces energy transfer and dissipation between the droplet and the substrate, thereby altering the dynamic characteristics of the impact process.^{23,24} Weisensee *et al.*²⁵ studied the impact of a droplet on a flexible superhydrophobic surface and found that the surface reduces the contact time by 50% through a “springboard effect” compared to a rigid surface. The mechanism is that the surface absorbs kinetic energy and vibrates when impacted, and during the droplet’s receding phase, the elastic potential energy is converted into the droplet’s vertical momentum, causing it to rebound in a pancake shape. Therefore, the focus of recent research has gradually shifted to droplet-impact flexible surfaces, which have great potential for applications.

Recently, researchers have studied a droplet impacting superhydrophobic fixed-fixed flexible surfaces and made some progress. For example, Ma *et al.*²⁶ studied the impact of a droplet on a superhydrophobic flexible surface through numerical simulation and found that the vertical movement of the surface reduces the maximum spreading diameter of the droplet compared to a rigid surface. Subsequently, researchers^{27,28} modeled the maximum spreading diameter of the droplet impacting on a flexible superhydrophobic cantilever wire mesh and on oscillating hydrophobic substrates. Hao *et al.*²⁹ simulated a droplet impacting a superhydrophobic fixed-fixed beam and found that the changes in We lead to the transition of the impact outcomes (spherical rebound, pancake rebound, and splash). They further pointed out the thresholds for the state transition (i.e., $We < 1.25$, $1.25 < We < 55.56$, and $We > 55.56$, respectively). Due to the vibration of the beam, the droplet rebounds off the surface earlier, reducing the contact time compared to a rigid beam. Xiong *et al.*³⁰ further investigated the droplet rebound by numerical analysis and proposed a control condition to promote the rebound and reduce the contact time, i.e., $We > 8$, $F_r \approx 1/T$. ($F_r = f_d/f_n$, f_n is the intrinsic frequency of the flexible surface, f_d is the vibration frequency of the droplet, and T is the contact time of the droplet impacting the rigid surface.) Under these circumstances, the bouncing takes place in the phase range of $(\pi, 2\pi)$. During the receding phase, the surface energy of the droplet and the

elastic energy of the surface convert into the kinetic energy simultaneously. The droplet can obtain more kinetic energy and larger upward acceleration before it bounces off the surface. Lee *et al.*³¹ studied the splash dynamics and found that the splash behavior is affected by We and surface stiffness; that is, in the case of high Weber number and low surface stiffness, splashing is more easily induced. Oleron *et al.*³² showed that energy dissipation partition between a liquid and a soft viscoelastic substrate controls the shape and stability of droplets sliding on soft viscoelastic substrates.

Previous studies have demonstrated that, in the case of droplet impacts on flexible substrates, the impact dynamics—including spreading, receding, and splashing behaviors—and the rebound characteristics of the droplet are primarily governed by the Weber number and the surface flexibility. However, quantitatively predicting the rebound behavior and determining the corresponding contact time in practical applications remains challenging, as the influence of substrate oscillation characteristics on rebound dynamics is not yet fully understood. Therefore, it is essential to conduct a comprehensive investigation into the rebound dynamics of droplets impacting flexible substrates and to develop predictive models for contact time. Unfortunately, such a relationship has not been reported in the existing literature. Moreover, research on substrate oscillation characteristics induced by droplet impact remains limited. Recent findings³³ indicate that the impact force induces substrate oscillations, and the resulting deflection amplitude is strongly correlated with the magnitude of the impact force. In addition, studies^{27,34} on the deflection amplitude and vibration period of substrates have primarily focused on cantilever beams or wire meshes, with limited attention given to fixed-fixed flexible substrates. Therefore, a systematic investigation into how droplet impact dynamics and substrate flexibility jointly influence the deflection amplitude and vibration period of flexible substrates remains necessary. Consequently, it is essential to comprehensively study the rebound dynamics and substrate oscillations when a droplet impacts a fixed-fixed flexible substrate, aiming to elucidate the mechanisms governing contact time under oscillating surface conditions and to establish predictive models for contact time.

Therefore, in this study, a water droplet impacting a superhydrophobic fixed-fixed flexible substrate (PDMS) is studied experimentally. The effects of the impact Weber number ($We = 7.06\text{--}70.57$) and substrate flexibility ($E = 0.45\text{--}1.40$ MPa) on the droplet’s impact dynamics and substrate’s oscillation dynamics are investigated, respectively. Based on the analysis of the interaction phase between the impacting droplet and the flexible substrate, this study indicates the mechanism of the droplet rebound and finally builds the relationship between the contact time of the droplet impacting a flexible substrate.

II. EXPERIMENTAL METHOD

A. Preparation of flexible substrate

First, the pre-prepared glass molds were ultrasonically cleaned with acetone, anhydrous ethanol, and de-ionized water in sequence for 10 min to remove substrate oils and particles. Next, the crosslinker and curing agent (Sylgard 184) were mixed in a 10:1 mass ratio and stirred at 500 rpm for 10 min using a magnetic stirrer.³⁵ This was followed by centrifugation at 2000 rpm for 5 min, which could be repeated until no bubbles were present. The PDMS was slowly poured into the treated molds, the oven was preheated to 160 °C, and the molds were placed in the oven and held for 30 min. The prepared Substrate 1 ($5 \times 2.5 \times$

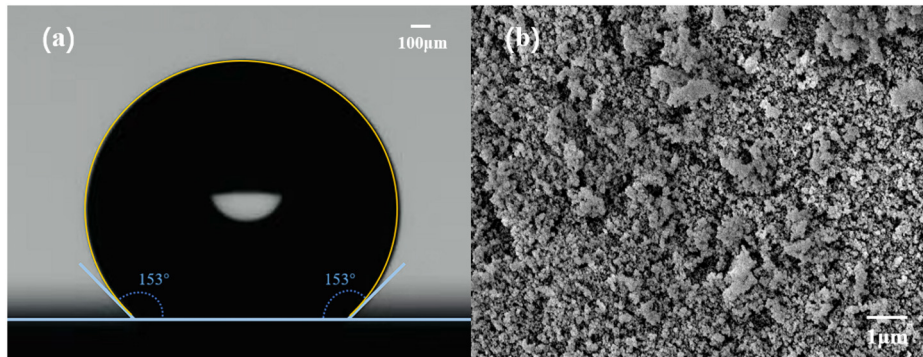


FIG. 1. (a) Static contact angle and (b) SEM images of the superhydrophobic substrate.

0.2 cm^3) was then washed sequentially with acetone, anhydrous ethanol, and de-ionized water, sprayed with a commercially available superhydrophobic coating (NeverWet Multisurface Aerosol).³⁶ The static contact angle of a de-ionized water droplet on the substrate that was measured multiple times as $\theta = 153^\circ \pm 1^\circ$ using a standard contact angle goniometer (DSA25S KPÜSS) is shown in Fig. 1(a). The micromorphology image of the superhydrophobic substrate was obtained using a scanning electron microscope (SEM, JCM-700, JEOL) in Fig. 1(b). Finally, the ratio was changed to 20:1, 30:1, and 40:1 in turn, and the above substrate preparation steps were repeated to obtain substrate 2, substrate 3, and substrate 4. The substrate parameters are shown in Table I.³⁷

B. Experiment system

The experimental setup is shown in Fig. 2. The experimental setup consists of an injection pumping system, a substrate fixation device, and a high-speed imaging system. The syringe pumping system controllably generates fixed-size droplets ($D_0 = 2.2 \text{ mm}$) from a 28G needle. A z-axis stage was used to adjust the height (i.e., the vertical distance between the needle and the substrate), and the impact velocity and Weber number varied according to the stage height. The substrate fixation device consisted of a fixed-fixed beam and a color laser coaxial displacement meter (CL-3000). The ends of the substrate were fixed, and the center was suspended to form a fixed-fixed beam structure, which is named as a flexible substrate in the following text. The surface vibration condition is measured using a color laser coaxial displacement meter to ensure that the surface is in a static state and free from external interference. The environmental parameters in the laboratory are closely monitored by an electronic hygrometer to ensure consistent thermal and acoustic conditions. Moreover, the temperature of the liquid is maintained at the room temperature level. The photograph acquisition system comprises a high-speed camera (ATX-I 100 mm

F2.8 with a maximum resolution of 1280×800 at 22 600 fps), a cold light (LED-100), and a light diffuser. The light diffuser was used to improve light distribution. This experimental camera uses a resolution of 640×480 to ensure a precise and smooth visualization of the droplet impact stage. The high-speed camera is connected to the computer as the data acquisition system, which not only enables precise control of the camera function but also ensures the safe storage of captured images. The camera operates at a frequency of 20 000 frames per second. The measurement of the contact time has an uncertainty of 0.05 ms, and the measurement of the contact diameter based on the pixel data of a single image has an uncertainty of 0.001 mm. The sampling frequency of the color laser coaxial displacement meter is 2000 Hz, with a measurement accuracy of 0.0001 mm, and the uncertainty of the oscillation deflection is less than 1%. Error analysis verified the reliability of the experimental data. To ensure the reliability and validity of the data, each experiment is repeated at least five times or more, and the average value is calculated.

III. RESULTS AND DISCUSSION

A. Influence of the impact Weber number

1. Droplet dynamics characteristics

The morphological evolution of a droplet impacting a superhydrophobic flexible substrate at different Weber numbers ($We = 7.06\text{--}70.57$) under certain substrate flexibility of $E = 0.62 \text{ N/m}$ and droplet diameters of $D_0 = 2.20 \text{ mm}$ is illustrated in Fig. 3. The droplet goes through spreading and receding phases after impacting the substrate. During the spreading phase, a part of the droplet's kinetic energy converts into its surface energy, where the droplet's diameter increases while its height reduces. Under the low Weber numbers ($We = 7.06\text{--}42.34$), the droplet undergoes deformation after impact, presenting a stacked state similar to a hamburger at around $t^* = 2.05 \text{ ms}$ in Fig. 3(a). Then the droplet continues to spread outward and reaches the maximum spreading diameter. At this time, the kinetic energy of the droplet tends to zero, the surface energy is maximum, and the center protrudes upwards. As the Weber number increases from $We = 7.06$ and $We = 42.34$, the spreading phenomenon is more intense, and the maximum spreading diameter gradually increases.

During the receding phase, part of the stored surface energy is converted into kinetic energy, and the fluid at the top continues to move upwards due to inertia, while the fluid at the bottom is slower due to friction with the substrate. This velocity gradient causes the droplet to stretch and form a neck. When the velocity difference

TABLE I. Details of the substrate in the present study.

Substrate	Proportion	E (MPa)
Substrate 1	10:1	1.40
Substrate 2	20:1	0.83
Substrate 3	30:1	0.62
Substrate 4	40:1	0.45

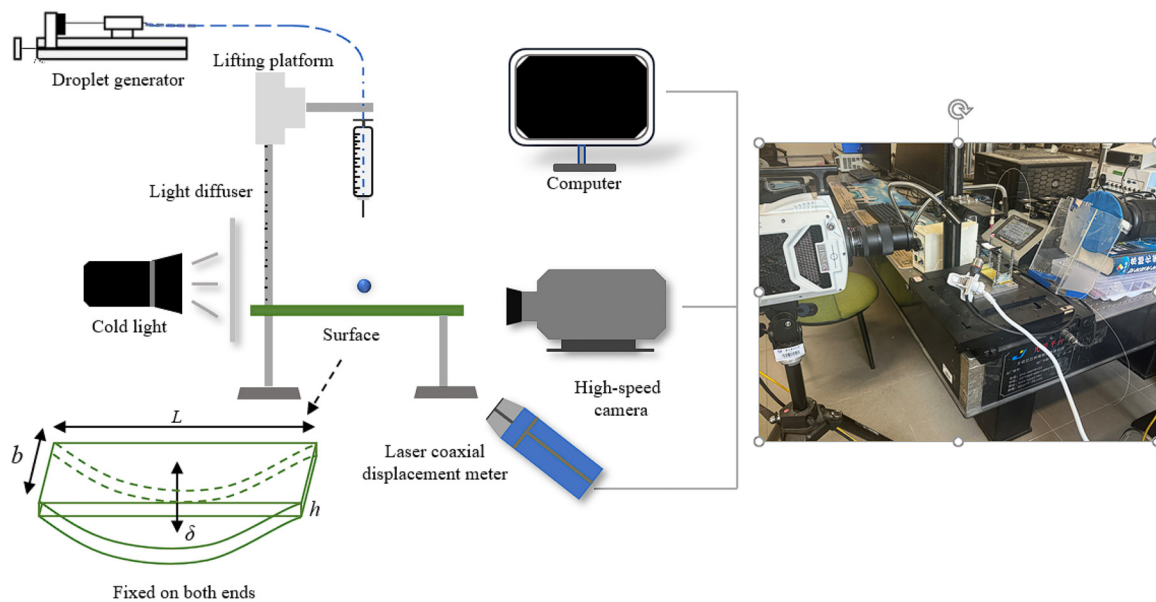


FIG. 2. Schematic diagram of experimental apparatus.

between the top and bottom exceeds what surface tension can regulate, the fluid may break off at the neck in Fig. 3(b). Under the high Weber numbers ($We = 42.34\text{--}70.57$) in Figs. 3(d)–3(f), the droplet takes on a pancake shape at the early phase of spreading. During the receding phase, the outer edge of the liquid layer is disturbed due to the effect of gas–liquid interfacial instability. In this case, the liquid film recedes more quickly, causing the droplet, which is not elongated, to detach from the substrate as a whole.

To quantitatively characterize the influence of Weber numbers on impact dynamics, Fig. 4 presents the variation of spreading diameter over time for four different levels of flexibility ($E = 0.45\text{--}1.40$) under varying Weber numbers ($We = 7.06\text{--}70.57$). The nondimensional spreading factor $\beta(t)$ is defined as $\beta(t) = D(t)/D_0$, where $D(t)$ denotes the contact diameter of the droplet on the substrate. During the spreading phase (see the orange area in Fig. 4), the maximum spreading diameter increases with the Weber number, while the spreading time remains constant. This indicates that the Weber number primarily affects the extent of droplet spreading, a phenomenon also observed in studies involving droplets impacting hydrophobic and superhydrophobic rigid substrates.^{38,39} The receding phase can be divided into two phases: an early phase and a late phase. In the early phase, the droplet's receding rate increases with the Weber number due to the simultaneous rise in maximum spreading factor and surface tension at the end of the spreading phase. This leads to a rapid contraction of the liquid film toward the center during the initial phase of receding. Surface tension dominates this early phase, promoting inertial receding, which is referred to as the inertial receding mode.⁴⁰ In contrast, during the late phase, the receding rate becomes independent of the Weber number. Here, the process is governed by capillary receding, where surface tension balances the residual inertial force, resulting in overlapping curves across different We values in Fig. 4. This observation aligns with

previous findings.^{41,42} For instance, Wang *et al.*⁴² reported that when a droplet impacts a hydrophobic surface, the receding rate in the inertial regime is proportional to the square root of the impact velocity. In the later phase of receding, kinetic energy is gradually dissipated, and surface tension and viscous forces become the dominant mechanisms. It should be noted that, in the early stage (see the green area in Fig. 4), the droplet undergoes “inertial retraction” under the influence of the inertia force and retains a well-defined rim-lamella structure (see the fourth column of Fig. 3), which is the core structure of the inertial retraction mode. In the late stage (see the purple area in Fig. 4), the droplet undergoes “capillary retraction” under the influence of capillary force, and the internal interfaces of the surrounding rim collide with each other and generate upward momentum (see the fifth column of Fig. 3), which is the core structural feature of the capillary retraction mode.

2. Substrate oscillation characteristics

When a droplet impacts a flexible substrate, the substrate may oscillate, leading to subsequent deformation, which is more easily observed at lower flexibility. The relationship between the vibration of the substrate deflection and time at different Weber numbers is shown in Fig. 5(a). As the Weber number increases, the substrate exhibits more violent vibrations, which can be attributed to the fact that a higher Weber number corresponds to greater initial kinetic energy that may be converted into the elastic potential energy of the substrate. As shown in Fig. 5(b), the maximum deflection scales linearly with the Weber number at a certain substrate flexibility. As the substrate flexibility decreases from 1.40 to 0.45 MPa, the slope of the corresponding straight line increases from 2.82×10^{-4} to 8.87×10^{-4} , which depends on the substrate flexibility, indicating the Weber number shows a more obvious effect on a smaller flexibility of the substrate.

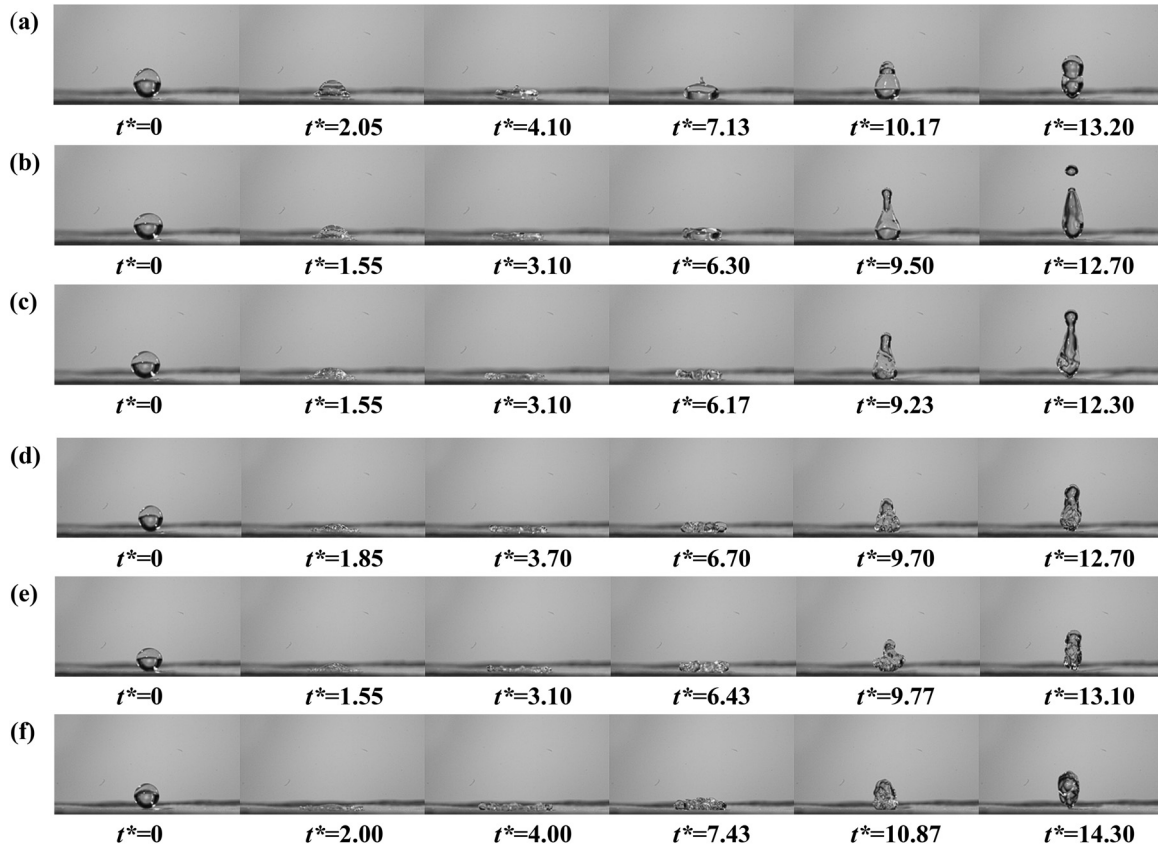


FIG. 3. Morphologic evolution of a water droplet impacting substrate with different Weber numbers: (a) $We = 7.06$; (b) $We = 21.17$; (c) $We = 35.28$; (d) $We = 42.34$; (e) $We = 56.45$; and (f) $We = 70.57$ and $E = 0.62$ MPa.

Furthermore, the Weber number has no discernible effect on the vibration period of the substrate in Fig. 5(a).

B. Influence of the flexibility of the substrate

1. Droplet dynamics characteristics

The morphological evolution of a droplet impacting a superhydrophobic flexible substrate at different flexibility ($E = 0.53$ – 1.40 MPa) and $We = 42.43$ is shown in Fig. 6. During the spreading phase, the droplet undergoes a transition from a straw hat to a flattened shape, and then reaches its maximum spreading diameter. It is worth noting that the maximum spreading diameter and spreading time are independent of the substrate flexibility in Fig. 7 because the inertial force mainly drives the spreading phase. Chen *et al.*⁴³ also found that the flexibility of a soft substrate does not affect the spreading dynamics. In other words, by the time the substrate is about to deform, the droplet has already completed its spreading phase, resulting in the effect of substrate flexibility being overshadowed by inertial forces. During the receding phase, the height of the liquid film decreases with the substrate flexibility, and finally, the droplet bounces off the substrate at almost the same time. As shown in Fig. 7, the substrate flexibility does not affect the receding rate and receding time. Alizadeh *et al.*⁴⁴ confirmed that droplet impact dynamics on superhydrophobic substrates

are independent of substrate flexibility. This phenomenon can be attributed to the characteristics of microstructured superhydrophobic surfaces in the Cassie state, where the contact area between the droplet and the substrate is limited to a small fraction of the microstructure tops. As a result, elastic deformation of the substrate cannot be effectively transferred to the droplet due to the presence of an intervening air layer. Previous studies^{42,45–47} have demonstrated that the actual contact area between the droplet and the superhydrophobic substrate is extremely small. Consequently, even if substrate deformation occurs, the localized strain within these minute contact regions is insufficient to influence the macroscopic receding behavior of the droplet.

2. Substrate oscillation characteristics

The variation of the fixed-fixed substrate deflection with time at different substrate flexibility, $E = 0.45$ – 1.40 MPa, and $We = 42.34$ is shown in Fig. 8. The substrate flexibility affects its deflection; that is, the deflection increases with the decrease in the flexibility. Additionally, the substrate flexibility also influences the vibration period of the substrate. This phenomenon can be interpreted as a substrate with lower flexibility undergoing greater deflection upon impact, and requiring a longer time to return to its original state. The lower the substrate flexibility, the more pronounced the effect on its vibration characteristics.

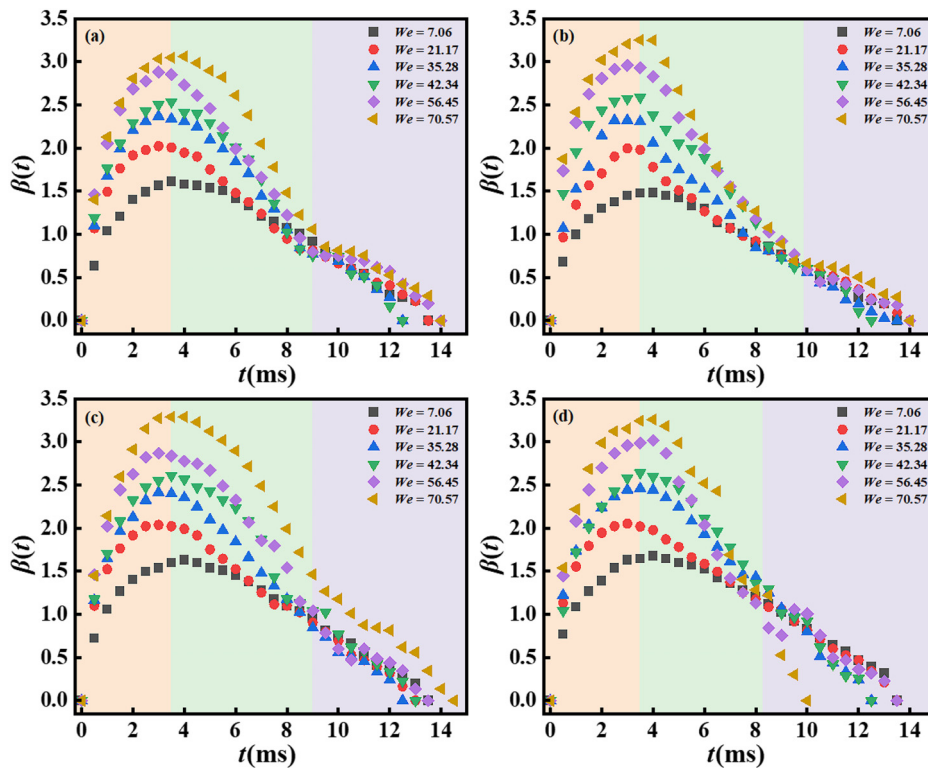


FIG. 4. Variation of the spreading factor with the impact time at different Weber numbers: (a) $E = 1.40$ MPa; (b) $E = 0.83$ MPa; (c) $E = 0.62$ MPa; and (d) $E = 0.45$ MPa. The spreading factor is defined as $\beta(t) = D(t)/D_0$. The orange area represents the spreading process, while the green and purple areas respectively represent the early and late stages of retraction.

C. Contact time of an impact droplet on a flexible substrate

The contact time of a droplet impacting a flexible superhydrophobic substrate, as a function of the Weber number ($We = 7.06$ – 70.57) and the substrate flexibility ($E = 0.45$ – 1.40 MPa), is presented in Fig. 9. The dimensionless contact time (t_c^*) is defined as the ratio of the contact time on a flexible superhydrophobic substrate (t_c) to that on a rigid superhydrophobic substrate (t_d), which can be expressed as $t_c^* = t_c/t_d$. While the contact time is influenced by the Weber number, it remains independent of the substrate flexibility. As the Weber number increases, the contact time initially decreases and subsequently increases, as illustrated in Fig. 9. This behavior contrasts with that

observed for droplets impacting a rigid superhydrophobic substrate, where the contact time is unaffected by the Weber number.

At lower Weber numbers ($We = 7.06$ – 42.04), the extent of droplet spreading upon impact is limited. As the Weber number increases, a greater proportion of kinetic energy is converted into surface energy, thereby enabling surface tension to drive rapid droplet recoil, resulting in a reduction in contact time. Fitting of the experimental data reveals that t_c^* and We follow a proportional relationship, which can be expressed as

$$t_c^* = -0.0023 We + 1.2156 \quad 7.06 < We < 42.04. \quad (1)$$

At higher Weber numbers ($We = 42.04$ – 70.57), the contact time increases with the Weber number. This phenomenon may be attributed

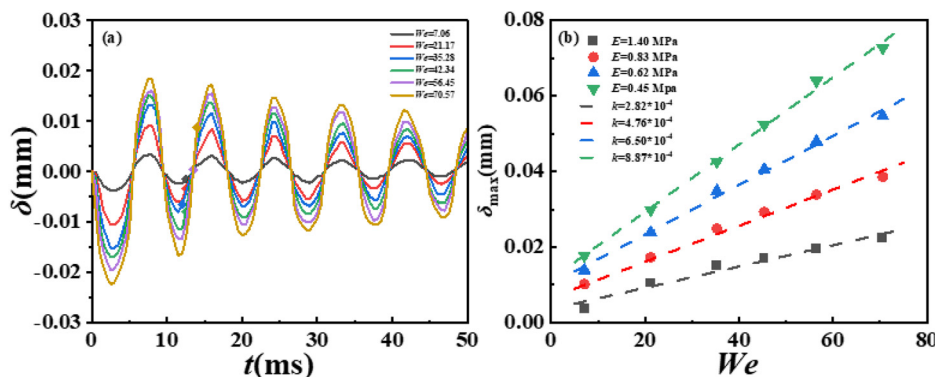


FIG. 5. (a) Variation of the substrate deflection with the impact time at different Weber numbers (7.06–70.57) of $E = 1.40$ MPa. (b) Variation of the maximum substrate deflection with different substrate flexibilities ($E = 0.45$ – 1.40 MPa).

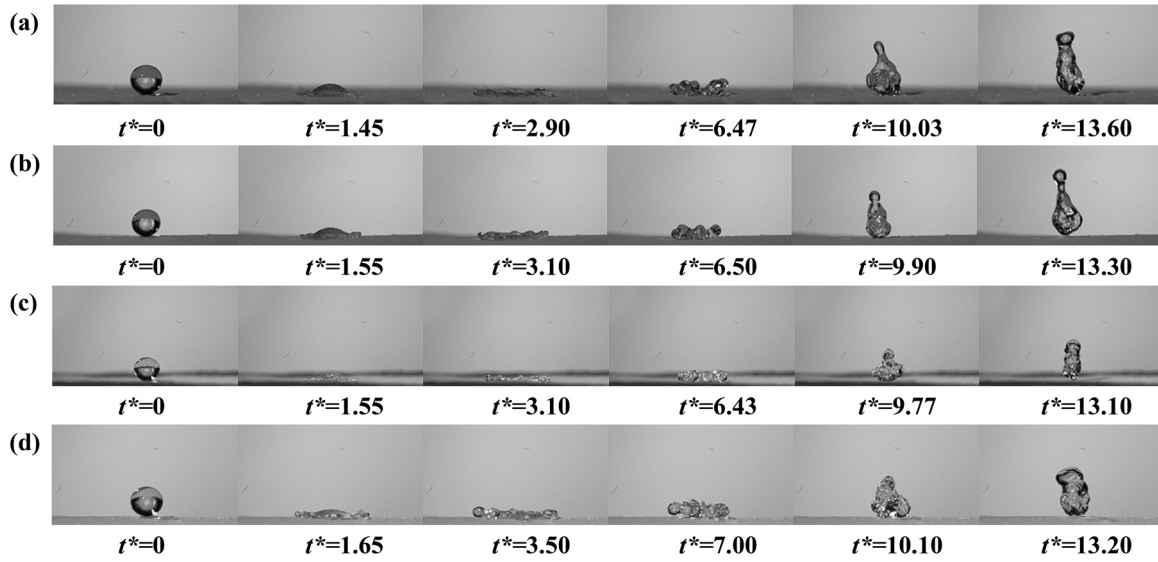


FIG. 6. Morphologic evolution of a water droplet impacting substrate with different substrate flexibility: (a) $E = 1.40$ MPa; (b) $E = 0.83$ MPa; (c) $E = 0.62$ MPa; and (d) $E = 0.45$ MPa and $We = 42.34$.

to the partial penetration of the micro- and nanostructures on the superhydrophobic substrate. When a droplet impacts with high initial kinetic energy, it may infiltrate the structural interstices, leading to partial destruction of the air layer and an increase in solid–liquid contact area. Consequently, the droplet recedes under greater resistance, significantly prolonging the contact time.

As shown in Fig. 9, the scaling law for t_c^* can be expressed as

$$t_c^* = 0.0049We + 0.9129 \quad 42.04 < We < 70.57. \quad (2)$$

Richard *et al.*¹⁶ proposed a theoretical scaling relationship for the contact time (t_d) of a droplet impacting a rigid superhydrophobic substrate, given by

$$t_d = 2.6(\rho R_0^3/\sigma)^{1/2}. \quad (3)$$

Based on this, the relationships in Eqs. (1) and (2) can be derived as follows:

$$t_c \sim We(\rho R_0^3/\sigma)^{1/2}. \quad (4)$$

By substituting $We = (\rho R_0 v_0^2)/\sigma$ into Eq. (4), the transformed expression becomes

$$t_c \sim \frac{R_0}{v} We^{3/2}. \quad (5)$$

Therefore, the scaling law for the contact time of a droplet impacting a flexible superhydrophobic substrate can be formulated as

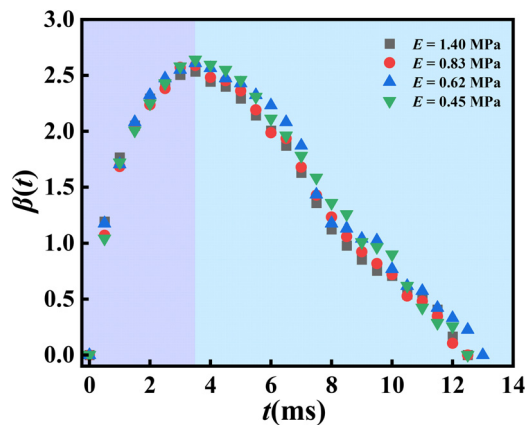


FIG. 7. Variation of the spreading factor with the impact time at different substrate flexibility ($E = 0.45$ – 1.40 MPa) and $We = 42.34$. The purple area represents the spreading process, while the blue area represents the retraction process.

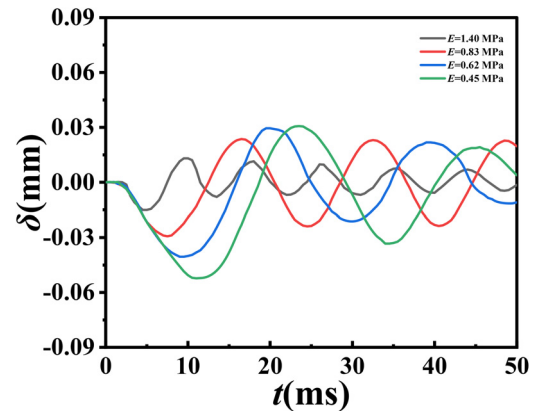


FIG. 8. Variation of the substrate deflection with time at different substrate flexibility ($E = 0.45$ – 1.40 MPa) and $We = 42.34$.

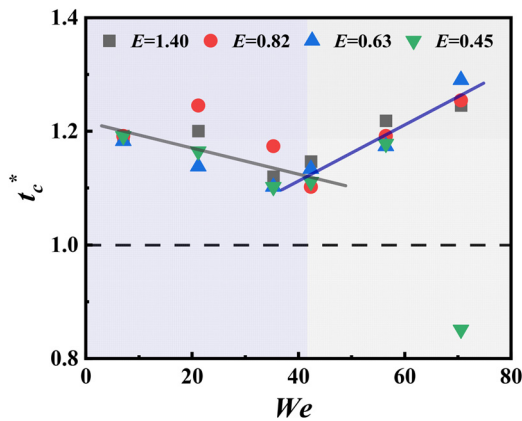


FIG. 9. Variation of the contact time with different Weber numbers ($We = 7.06$ – 70.57) and substrate flexibility ($E = 0.45$ – 1.40 MPa). The dimensionless contact time is defined as $t_c^* = t_c/t_d$. For the droplet with an initial diameter of $D_0 = 2.2$ mm in this study, the theoretical contact time is expressed by the dotted line.

$$t_c \sim k \frac{R_0}{\nu} We^{3/2} \begin{cases} (k < 0) & 7.06 < We < 42.04, \\ (k > 0) & 42.04 < We < 70.57. \end{cases} \quad (6)$$

However, the contact time observed on a flexible substrate is longer than that on the rigid substrate. This difference may be attributed to the influence of substrate vibrations on the impact dynamics. Upon droplet impact, substrate vibrations convert part of the kinetic energy into elastic potential energy, contributing to energy dissipation and

thus increasing the contact time. Figure 10 presents a schematic of the substrate motion during droplet rebound. The droplet detaches during the upward movement of the vibrating substrate. Since the droplet moves in the same direction as the substrate at this phase, the relative velocity between them decreases, further contributing to the prolonged contact time.

The research results indicate that most of the experimental data points closely match the scaling line shown in Fig. 9. Within a wide parameter space of Weber numbers ($We = 7.06$ – 70.57) and substrate flexibility ($E = 0.45$ – 1.40 MPa), Eq. (6) has reliable predictive performance for the contact time of droplets impacting on the flexible superhydrophobic substrates at normal temperature. It should be noted that this correlation was specifically verified for low-viscosity fluids (for example, in this study, the dynamic viscosity was approximately $\mu = 1.0$ mPa s of de-ionized water). Future research should focus on expanding the application scope of the model, especially exploring its performance in high-viscosity fluid systems and suspensions of droplets containing particles.

IV. CONCLUSION

The coupled dynamics of a droplet impacting flexible superhydrophobic substrates are experimentally investigated with high-speed imaging and position sensing at different Weber numbers ($We = 7.06$ – 70.57) and substrate flexibility ($E = 0.45$ – 1.40 MPa) in this study. The findings elucidate the coupled effects between droplet impact dynamics and surface oscillation characteristics. For droplet's impact dynamics, the spreading phase is not affected by the We number and substrate flexibility since the inertial action is dominant and the action of the flexible substrate is relatively small, while the receding

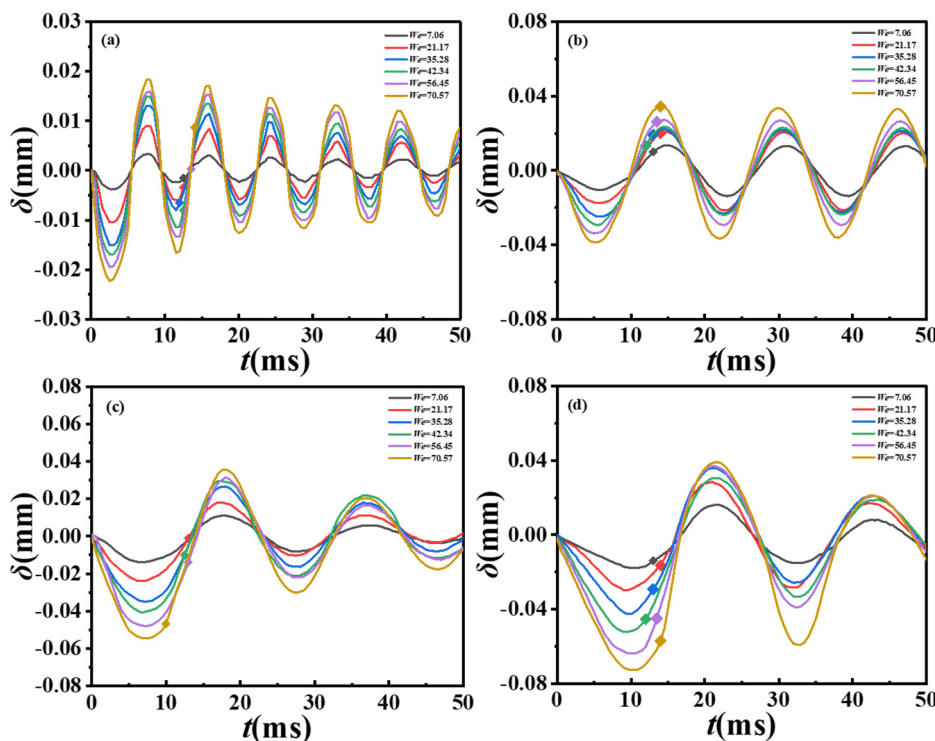


FIG. 10. Variation of the substrate deflection with time at different substrate flexibility of (a) $E = 1.40$ MPa, (b) $E = 0.83$ MPa, (c) $E = 0.62$ MPa, and (d) $E = 0.45$ MPa.

phase is closely related to the We number. For flexible substrate's oscillation dynamics, the vibrating deflection increases with the increase in the We number and reduction of the substrate flexibility, while the vibration period of the substrate is only dependent on its flexibility. The contact time observed on a flexible substrate is longer than that on a rigid substrate, which may be attributed to the scale relationship between the droplet and substrate oscillation period. Based on these findings, a relationship for the contact time of a droplet impacting fixed-fixed flexible superhydrophobic substrates is established for the first time. At lower Weber numbers ($We = 7.06\text{--}42.04$), the relationship is expressed as $t_c^* = -0.0023We + 1.2156$; at higher Weber numbers ($We = 42.04\text{--}70.57$), it is given by $t_c^* = 0.0049We + 0.9129$. This study systematically investigates the droplet impact dynamics and surface oscillation characteristics during droplet impingement on a fixed-fixed flexible substrate, thereby contributing to the quantitative prediction of rebound behavior and the determination of the corresponding contact time.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Shu-Rong Gao: Writing – original draft (equal); Writing – review & editing (equal). **Yu-Xiang Wang:** Investigation (equal); Methodology (equal); Writing – original draft (equal). **Lian-Kai Shi:** Investigation (equal). **Shao-Fei Zheng:** Investigation (equal). **Yan-Ru Yang:** Investigation (equal). **Xiao-Dong Wang:** Funding acquisition (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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